

CONTACT

EMAIL

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PHONE

(646) 470-4361

LINKEDIN

linkedin.com/in/valpetrov

LOCATION

New York, NY

EXPERTISE

- **AI & Agentic**
LLM Agents
Multi-Agent Orchestration
RAG Pipelines
Tool Use / Function Calling
Prompt Engineering
LangChain / LangGraph
Vector Databases
- **LLM Platforms**
OpenAI API
Claude / Anthropic API
Hugging Face
Pinecone · ChromaDB
- **Languages**
Python · Kotlin · Java
TypeScript · C# · SQL
- **Infrastructure & Data**
Apache Kafka · Databricks
Redis · SQL Server
Hadoop / HDFS · Delta Lake
- **Domain Background**
FinTech / Wall Street
Security Master
Portfolio Management
Real-Time Systems
- **Leadership**
Team Building
Systems Architecture
Agile · Hiring · Mentoring

EDUCATION

B.S. Computer Eng.

University of Waterloo

Ontario · Class of 2008

GPA: 3.7 / 4.0

SNAPSHOT

14+ Years in distributed systems

5 Tier-1 Wall Street firms

8+ Engineers led on PMS

6M+ Txns/day at sub-sec latency

AGENTIC ENGINEER · NEW YORK, NY

Val Petrov

Building autonomous *agents* that move industries.

PROFILE

Agentic Engineer with 14+ years architecting mission-critical distributed systems for top-tier Wall Street firms — now focused on autonomous AI agents, LLM-powered workflows, and multi-agent orchestration. Deep expertise across real-time data pipelines, platform architecture, and team leadership. I build systems — and agents — that reason, act, and ship.

EXPERIENCE

• APR 2025 — PRESENT

Tech Lead — Macro Technology

Point72 · New York

Leading technology for the macro investment platform. Orchestrating Murex migration with end-to-end ownership of trade booking, straight-through processing, and reconciliation across a distributed financial data stack.

Murex C# .NET Python 3.14 SQL Server Sybase Kafka Databricks

• NOV 2022 — JAN 2025

Data Warehouse Lead — Security Master

Centiva Capital · New York

Designed and built the firm's new security master from the ground up. Architected authorization services, instrument creation pipelines, and vendor data integrations (Bloomberg Sanctions, Corporate Actions, Pricing). Led sprints, hiring, and onboarding.

Python 3.11 SQL Server Databricks Git

• JUL 2018 — NOV 2022

Software Engineering Lead — Portfolio Management

Schonfeld · New York

Led a team of 8 engineers delivering an in-house PMS processing 6M+ transactions/day with sub-second latency. Built security master validation pipelines and a scalable Kafka-based architecture. Spearheaded Agile practices and mentored the team.

Kotlin Java 8/11 Kafka MySQL Redis kdb+

• JAN 2017 — JUL 2018

Senior Software Developer — Middle Office

Millennium Partners · New York

Developed invoicing, expense, and document management systems. Consolidated T+1 PMS PnL reporting; implemented a Kafka & Python POC for T-0 PnL delivery; reduced redundant Murex trade data by 88% via SQL optimization.

C# 7 .NET 4.0 Kafka Python RabbitMQ

• MAR 2015 — MAY 2016

Senior Software Developer — Big Data Platform

Bloomberg L.P. · New York

Built RESTful services for file storage and retrieval. Architected Big Data solutions using Hadoop HDFS and Solr handling hundreds of millions of messages daily. Engineered schema-driven extraction and search services.

Java Hadoop HBase Solr Python Flask

• MAR 2011 — MAR 2015

Senior Software Developer — MBS Structuring & Analytics

Bloomberg L.P. · New York

Built domain-rich GUI applications for Mortgage-Backed Securities structuring using .NET and WPF. Developed distributed server-side services, authored custom controls, and invented a novel concurrency control solution for improved throughput.

C# WPF .NET C++

• SEP 2010 — MAR 2011

Software Engineer — Director

Trading Machines · Stamford, CT

Led front-end GUI development for 90+ trading forms across 7+ applications. Managed multicast traffic consistency and built proprietary libraries for back-office functions. Coordinated market data feeds from OPRA, CTA, UTP, and direct exchanges.

C# WinForms C++ Perl FIX Protocol